
SOFTWARE REQUIREMENTS SPECIFICATION

for

BookHaven

Version 1.0

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1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for BookHaven, Version 1.0, a web-based online bookstore e-commerce platform being built from scratch. BookHaven allows sellers to create accounts and list books for sale, allows buyers to search, browse, and purchase those books, and allows an administrator to review and moderate listings before they are published. The scope of this document covers the full BookHaven system as planned for the initial release: account management for buyers, sellers, and administrators; book-listing creation, review, and approval; search and browse; shopping cart and checkout; order processing and payment; and post-purchase returns/refunds. It does not cover the internal implementation of the third-party payment gateway, hosting/deployment infrastructure, or a future native mobile application, which are outside the scope of this release.

1.2. Document Conventions

This document follows the IEEE 830 SRS template conventions. Chapters and sections are numbered hierarchically (e.g., 4.1, 4.1.1) to support cross-referencing between requirements. Individual functional requirements are labeled REQ-*n* in the order they appear within each System Feature section in Chapter 3; requirement priority (High, Medium, or Low) is stated once at the feature level and is inherited by every requirement listed under that feature unless a requirement explicitly states its own priority. Text in angle brackets (e.g., <TBD>) marks information that has not yet been finalized by the team.

1.3. Intended Audience and Reading

This SRS is intended for the BookHaven development team, QA testers, course instructors/graders, and any technical writers supporting the project. Developers and testers should focus on Chapter 3 (System Features) and Chapter 4 (Other Nonfunctional Requirements), which define the concrete functional and quality requirements the system must satisfy. Instructors or stakeholders evaluating the project should begin with Chapter 1 (Introduction) and Chapter 2 (Overall Description) to understand the product's purpose, scope, and intended users before reviewing the detailed requirements in the chapters that follow. Readers interested in the traceability between requirements and elicited needs should consult the Appendix, which lists the user stories and acceptance

criteria that every functional requirement in Chapter 3 was derived from. Readers should proceed sequentially, since later chapters build on the context established in Chapters 1 and 2.

1.4. Product Scope

BookHaven is a web-based e-commerce bookstore that connects independent booksellers with readers. Sellers create accounts, list books with a title, author, ISBN, genre, condition (new or used), description, cover image, price, and stock level, and manage their own inventory. Buyers search and browse book listings by title, author, genre, or ISBN, view book details, add books to a cart, and complete checkout with payment processing. An administrator reviews new book listings before they go live, screening for prohibited or inappropriate content and verifying listing details (such as ISBN format) so that only appropriate, accurate listings are published. Buyers may also request a return or refund on an eligible order. The objective of BookHaven is to give small and independent booksellers a simple, low-cost way to reach readers online without building their own storefront, while giving the platform operator control over listing quality, content, and trust.

1.5. Definitions, Acronyms, and Abbreviations

- **Buyer:** A registered user of BookHaven who searches, browses, and purchases book listings.
- **Seller:** A registered user of BookHaven who creates and manages book listings for sale.
- **Administrator (Admin):** A platform staff member with elevated privileges who reviews pending listings and enforces content policy.
- **Listing:** A single book entry created by a Seller, consisting of a title, author, ISBN, genre, condition, description, price, stock quantity, and optional cover image.
- **Pending Review:** The state of a listing that has been submitted or edited by a Seller but not yet approved or rejected by an Administrator.
- **Published:** The state of a listing that has been approved by an Administrator and is visible to Buyers in search and browse results.
- **ISBN:** International Standard Book Number; a unique numeric identifier for a book, validated against the ISBN-10 or ISBN-13 checksum format.
- **Cart:** A temporary collection of listings a Buyer intends to purchase, held until checkout is completed or the items are removed.

- **Checkout:** The process by which a Buyer supplies a shipping address and payment information to complete the purchase of the items in their cart.
- **Payment Gateway:** A third-party service that processes payment transactions on behalf of BookHaven without BookHaven storing raw payment card data.
- **Order:** A record created upon successful payment, containing the purchased listing(s), the Buyer, and the transaction details.
- **Return/Refund:** A post-purchase process by which a Buyer requests, and may receive, reimbursement for an eligible completed order.
- **Wishlist:** A personal, non-cart collection of listings a Buyer has saved for future reference or purchase.
- **PCI-DSS:** Payment Card Industry Data Security Standard; a set of security requirements for organizations that handle payment card information.

1.6. References

- [1] IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- [2] K. E. Wiegers, Software Requirements Specification template, http://www.se.rit.edu/~emad/teaching/slides/srs_template_sep14.pdf, the source template that this document's chapter/section structure is based on.
- [3] J.-P. Eisenbarth, SRS-TeX: A LaTeX template for a Software Requirements Specification respecting IEEE standards, <https://github.com/jpeisenbarth/SRS-TeX>, the LaTeX implementation of [2] used as the starting point for this file.
- [4] BookHaven User Stories and Use Cases, Cindy Cardona-Felix, Rakshit Jaiswal, Jalil Jimenez, Everett Wappler, see Appendix A of this document.
- [5] Cindy Cardona-Felix, Rakshit Jaiswal, Jalil Jimenez, Everett Wappler, "BookHaven Project Repository," <https://github.com/Raksj12/Software-Engineering-Project>, 2026.

2. Overall Description

2.1. Product Perspective

BookHaven is a new, self-contained web application being developed from scratch; it is not a follow-on to, or replacement for, any existing system. It consists of a front-end web client, a back-end application server, and a database that stores user accounts, book listings, and orders. The system interfaces with an external payment gateway to process transactions and, optionally, with an email/notification service to alert sellers and buyers about listing approvals, rejections, order confirmations, and return/refund updates. *<Insert a diagram here showing the Buyer Client, Seller Client, Admin Client, Application Server, Database, and external Payment Gateway, and how they connect.>*

2.2. Product Functions

BookHaven must let users perform the following major functions, detailed further in Chapter 3:

- Register, log in, and manage a buyer, seller, or admin account
- As a seller, create, edit, and remove book listings, including title, author, ISBN, genre, condition, description, cover image, price, and stock
- As an admin, review pending book listings and approve or reject them, including screening for prohibited content and verifying listing details
- As a buyer, search and browse published book listings by title, author, genre, or ISBN, and view book details
- Add books to a cart and complete checkout, including payment processing
- Receive notifications/confirmations for order placement, listing approval, and listing rejection
- As a buyer, request a return or refund on an eligible order

2.3. User Classes and Characteristics

BookHaven has three user classes. **Buyers** are readers who search, browse, and purchase books; they require no special technical expertise, only basic web-browsing familiarity. **Sellers** are registered booksellers (individuals or small shops) who list, price, and manage

their own book inventory; they need moderate familiarity with the platform’s listing tools but no technical expertise. **Administrators** are platform staff with elevated privileges who review pending book listings, approve or reject them, and enforce content policy; they are expected to understand the platform’s moderation rules in detail. Buyers and sellers are the most important user classes, since they drive core bookstore activity; administrators are a smaller but critical class responsible for maintaining listing quality and trust.

2.4. Operating Environment

BookHaven operates as a web application accessed through modern desktop and mobile browsers (current and previous major versions of Chrome, Firefox, Safari, and Edge), with no client-side installation required beyond a browser with JavaScript enabled and an internet connection. For Sprint 1 development and testing, the back end runs on `localhost` on each developer’s machine, using Node.js with the Express framework and a local MySQL database instance; this allows the team to build and test features without depending on shared server infrastructure during early development. A production deployment environment (e.g., a hosted Linux server or cloud platform) will be selected in a later sprint once the core application is functional. Regardless of environment, the back end must coexist with the third-party payment gateway’s client-side scripts and, if used, an email/notification service.

2.5. Design and Implementation Constraints

BookHaven must integrate with a third-party payment processor and must not store raw payment card data directly, in line with PCI-DSS considerations. Uploaded book cover images are restricted to approved file types and a maximum file size to control storage cost and prevent abuse. Listings must be checked against a maintained blacklist of prohibited words and phrases before publication, and ISBNs must be validated against the standard ISBN-10/ISBN-13 checksum format before a listing can be published. The system must support concurrent access by multiple sellers and buyers without data conflicts, such as overselling an out-of-stock book. Development is constrained to the technologies, tools, and timeline set by the course requirements, and the delivered system must be maintainable by the current development team without proprietary tooling.

2.6. Assumptions and Dependencies

It is assumed that buyers and sellers have access to a computer or mobile device with a modern web browser and a working internet connection. It is assumed that sellers will provide accurate information (title, author, ISBN, condition) when creating a listing, and that the admin review step will catch and reject any listing that is inaccurate, prohibited, or fraudulent. The system depends on the continued availability and correctness of the third-party payment gateway to process transactions; if the payment gateway is

unavailable, checkout cannot be completed. The system also depends on a reliable ISBN validation source or algorithm being available at listing-creation time. It is assumed that a single administrator or small administrative team will be sufficient to review listings for the initial release, and that this workload will not exceed what can reasonably be handled without additional automation.

3. System Features

3.1. User Account Management

3.1.1. Description and priority

This feature allows a person to register as a buyer or seller, log in and out, and manage basic account information (name, email, password, and, for sellers, a store/display name). Administrator accounts are created directly by the platform operator rather than through public self-registration. This feature is foundational to the rest of the system, since nearly every other feature requires the user to be authenticated first. Priority: **High**.

3.1.2. Stimulus/Response Sequences

- User navigates to the registration page, selects account type (Buyer or Seller), and submits name, email, and password → system creates the account and redirects the user to their home page, now logged in.
- User submits registration information that is incomplete or uses an email already on file → system displays an inline error identifying the missing or conflicting field and does not create the account.
- User submits a valid email and password on the login page → system authenticates the user and redirects to their role-appropriate home page (Buyer, Seller, or Admin).
- User submits an incorrect email/password combination → system displays an error message and offers a “forgot password” option.
- Logged-in user selects “Log Out” → system ends the session and redirects to the public landing page.

3.1.3. Functional Requirements

REQ-1: The system shall allow a new user to register as either a Buyer or a Seller by supplying a name, a unique email address, and a password.

REQ-2: The system shall reject a registration attempt that omits a required field or supplies an email address already associated with an existing account, and shall display a specific error message identifying the problem.

REQ-3: The system shall allow a registered user to log in using their email and password,

and shall redirect the user to the home page appropriate to their role (Buyer, Seller, or Admin).

REQ-4: The system shall display an error message when login credentials do not match an existing account, without revealing whether the email or the password was incorrect.

REQ-5: The system shall allow a logged-in user to log out, ending their session.

REQ-6: The system shall allow a logged-in user to view and edit their own account information (name, email, password, and store name for sellers), except that account type (Buyer/Seller/Admin) shall not be user-editable after registration.

REQ-7: The system shall automatically end a user's session after 30 minutes of inactivity, requiring the user to log in again to continue.

3.2. Book Listing Management

3.2.1. Description and priority

This feature allows a Seller to create, edit, and remove book listings in their own inventory. A listing includes a title, author, ISBN, genre, condition (new or used), description, cover image, price, and stock quantity. Newly created and edited listings are not visible to buyers until they have passed admin review (see Section 3.3). Priority: **High**.

3.2.2. Stimulus/Response Sequences

- Seller selects “Add Listing” and submits title, author, ISBN, genre, condition, description, price, and stock, optionally with a cover image → system validates the ISBN format, creates the listing in a “Pending Review” state, and adds it to the admin review queue.
- Seller submits a listing with a malformed ISBN → system rejects the submission and displays an error indicating the ISBN is invalid.
- Seller selects “Edit” on an existing listing and changes one or more fields → system saves the changes and, if a field relevant to moderation (title, description, ISBN) was changed, returns the listing to “Pending Review.”
- Seller selects “Remove” on a listing → system asks for confirmation, then removes the listing from the store (published or not) upon confirmation.
- Seller updates the stock quantity of a listing to zero → system marks the listing as “Out of Stock” and prevents buyers from adding it to their cart until stock is replenished.

3.2.3. Functional Requirements

REQ-8: The system shall allow a Seller to create a new book listing with a title, author, ISBN, genre, condition (new or used), description, price, and stock quantity, and an optional cover image.

REQ-9: The system shall validate that a submitted ISBN conforms to the ISBN-10 or ISBN-13 checksum format before accepting a listing, and shall reject the submission with an error message otherwise.

REQ-10: The system shall place every newly created or materially edited listing into a “Pending Review” state that is not visible to Buyers until an Administrator approves it.

REQ-11: The system shall allow a Seller to edit the fields of their own existing listings.

REQ-12: The system shall allow a Seller to remove their own listing from the store at any time, regardless of its review status.

REQ-13: The system shall prevent Buyers from adding a listing with zero stock to their cart, and shall visibly label such listings as “Out of Stock.”

REQ-14: The system shall restrict a Seller’s edit and remove actions to listings that they themselves created.

3.3. Listing Review and Approval

3.3.1. Description and priority

This feature allows an Administrator to review book listings submitted by sellers before they are published, screening for prohibited or inappropriate content and verifying that listing details (particularly the ISBN) are valid, so that only appropriate and accurate listings reach buyers. Priority: **High**.

3.3.2. Stimulus/Response Sequences

- Administrator opens the review queue → system displays all listings currently in the “Pending Review” state, in order of submission.
- Administrator selects a pending listing and clicks “Approve” → system marks the listing as “Published,” making it visible to Buyers in search and browse, and notifies the Seller of the approval.
- Administrator selects a pending listing and clicks “Reject,” entering a reason → system marks the listing as “Rejected,” keeps it hidden from Buyers, and notifies the Seller with the stated reason.
- Administrator opens a pending listing whose title or description matches a prohibited-content pattern → system flags the listing with a warning indicator before the Administrator decides.

3.3.3. Functional Requirements

REQ-15: The system shall present Administrators with a queue of all listings in the “Pending Review” state, ordered by submission date.

REQ-16: The system shall allow an Administrator to approve a pending listing, which shall change its state to “Published” and make it visible to Buyers.

REQ-17: The system shall allow an Administrator to reject a pending listing along with a reason, which shall change its state to “Rejected” and keep it hidden from Buyers.

REQ-18: The system shall notify the Seller by email or in-app notification whenever one of their listings is approved or rejected, including the reason in the case of a rejection.

REQ-19: The system shall automatically flag a pending listing for Administrator attention if its title or description matches an entry in a maintained list of prohibited words or phrases.

REQ-20: The system shall re-validate the ISBN format of a listing at review time and shall prevent an Administrator from approving a listing with an invalid ISBN.

3.4. Search and Browse

3.4.1. Description and priority

This feature allows a Buyer to find books by searching or browsing published listings. Buyers can search by title, author, genre, or ISBN, and can browse listings organized by genre. Priority: **High**.

3.4.2. Stimulus/Response Sequences

- Buyer enters a search term and selects a search field (Title, Author, Genre, or ISBN) → system displays all Published listings matching the term in the selected field, sorted alphabetically by title.
- Buyer’s search term matches no published listings → system displays a “no results found” message.
- Buyer selects a genre from the browse menu → system displays all Published listings in that genre.
- Buyer selects a specific listing from search results or browse results → system displays the full listing detail page, including description, price, condition, seller name, and stock availability.

3.4.3. Functional Requirements

REQ-21: The system shall allow a Buyer to search Published listings by title, author, genre, or ISBN.

REQ-22: The system shall display search results sorted in ascending alphabetical order by title.

REQ-23: The system shall display a clear “no results found” message when a search returns no matching Published listings.

REQ-24: The system shall allow a Buyer to browse Published listings grouped by genre without entering a search term.

REQ-25: The system shall display, for any selected listing, its title, author, ISBN, genre,

condition, description, cover image (if provided), price, seller name, and current stock availability.

REQ-26: The system shall exclude listings that are not in the “Published” state from all search and browse results.

3.5. Shopping Cart and Checkout

3.5.1. Description and priority

This feature allows a Buyer to add published, in-stock listings to a shopping cart and complete a purchase, including payment processing through a third-party payment gateway. Priority: **High**.

3.5.2. Stimulus/Response Sequences

- Buyer clicks “Add to Cart” on a Published, in-stock listing → system adds the item to the Buyer’s cart and updates the cart item count and subtotal.
- Buyer views their cart and clicks “Remove” on an item → system removes the item and recalculates the subtotal.
- Buyer with at least one item in their cart clicks “Checkout” while not logged in → system prompts the Buyer to log in or register before proceeding.
- Logged-in Buyer clicks “Checkout” → system displays an order summary and a payment form.
- Buyer submits valid payment information → system processes the payment through the payment gateway, decrements stock for each purchased item, creates an order record, sends a confirmation, and displays a thank-you page.
- Buyer submits payment information that the gateway declines → system displays an error message and returns the Buyer to the payment form without modifying stock or creating an order.

3.5.3. Functional Requirements

REQ-27: The system shall allow a Buyer to add a Published, in-stock listing to their shopping cart.

REQ-28: The system shall allow a Buyer to view their cart, remove items from it, and see an updated subtotal after any change.

REQ-29: The system shall require a Buyer to be logged in before proceeding from the cart to checkout, prompting login or registration if they are not.

REQ-30: The system shall display an order summary, including itemized prices and the total cost, before collecting payment information.

REQ-31: The system shall process payment through the third-party payment gateway

and shall not store raw payment card information on BookHaven's own servers.

REQ-32: The system shall, upon a successful payment, decrement the stock quantity of each purchased listing, create a permanent order record, and send an order confirmation to the Buyer.

REQ-33: The system shall, upon a declined or failed payment, leave stock quantities and order records unchanged and shall display an error message allowing the Buyer to retry.

3.6. Returns and Refunds

3.6.1. Description and priority

This feature allows a Buyer to request a return or refund on an eligible completed order, and allows the request to be processed against the platform's return policy. Priority: **Medium**.

3.6.2. Stimulus/Response Sequences

- Buyer views their order history and selects "Request Return" on an eligible order → system displays a form for the Buyer to state a reason for the return.
- Buyer submits a return request within the eligible window (e.g., 30 days of purchase) → system creates a return request record, notifies the Seller, and displays a confirmation to the Buyer.
- Buyer attempts to request a return on an order outside the eligible window → system displays a message explaining that the order is no longer eligible for return.
- Seller or Administrator approves a submitted return request → system marks the return as approved, issues a refund through the payment gateway, and notifies the Buyer.

3.6.3. Functional Requirements

REQ-34: The system shall allow a Buyer to request a return on a completed order within a configurable eligibility window from the purchase date.

REQ-35: The system shall reject return requests submitted after the eligibility window has passed and shall inform the Buyer of the reason.

REQ-36: The system shall require a Buyer to provide a reason when submitting a return request.

REQ-37: The system shall notify the relevant Seller when a return request is submitted against one of their sold listings.

REQ-38: The system shall allow an approved return request to trigger a refund through the payment gateway and shall update the order's status to "Refunded."

REQ-39: The system shall notify the Buyer when their return request is approved or denied.

3.7. Order Fulfillment

3.7.1. Description and priority

This feature allows a Seller to process, ship, and track orders placed against their listings once payment has succeeded, and allows the Seller to cancel an order they cannot fulfill. For digital listings, this feature also covers automatic delivery of a license or download link without Seller intervention. Priority: **High**.

3.7.2. Stimulus/Response Sequences

- Seller opens “Pending Orders” and selects an order → system displays the order details, including the purchased item(s), buyer shipping address, and payment status.
- Seller marks a physical order as “Processing,” selects a shipping carrier, and enters tracking information → system updates the order status, stores the tracking number, and notifies the Buyer that the order has shipped.
- Seller selects “Cancel Order” on a physical order and enters a cancellation reason → system cancels the order, issues a refund to the Buyer’s original payment method, and notifies the Buyer with the stated reason.
- System detects a confirmed payment on a digital listing → system automatically generates a unique access token or license tied to the Buyer’s account, delivers the file or link, and notifies the Seller that the order was fulfilled automatically, without requiring Seller action.

3.7.3. Functional Requirements

REQ-40: The system shall display, for each order in “Pending Orders,” the purchased item(s), Buyer shipping address, and payment status, restricted to orders containing that Seller’s own listings.

REQ-41: The system shall allow a Seller to mark a physical order as “Processing” and to record a shipping carrier and tracking number against that order.

REQ-42: The system shall notify the Buyer with the carrier and tracking number whenever a Seller marks their order as shipped.

REQ-43: The system shall allow a Seller to cancel a physical order that has not yet shipped, requiring a stated reason, and shall automatically issue a refund to the Buyer’s original payment method upon cancellation.

REQ-44: The system shall notify the Buyer whenever their order is canceled, including the Seller’s stated reason.

REQ-45: The system shall, upon confirmed payment for a digital listing, automatically generate a unique access token or license bound to the Buyer’s account and deliver the associated file or link without requiring Seller action.

REQ-46: The system shall notify the Seller that a digital order was fulfilled automatically, for their records.

3.8. Reviews and Ratings

3.8.1. Description and priority

This feature allows a Buyer whose order for a given book has been delivered to leave a single star rating and written review of that book, and allows other Buyers to read reviews when viewing a listing. Submitted reviews are automatically screened for prohibited content before they are published. Priority: **Medium**.

3.8.2. Stimulus/Response Sequences

- Buyer opens a delivered order for a book they have purchased and selects “Write a Review” → system displays a form for a star rating and a written comment.
- Buyer submits a rating and review → system checks the review for prohibited content, saves it if it passes, associates it with the book and the Buyer, and displays it in the reviews section of that book’s listing.
- Buyer attempts to submit a second review for a book they have already reviewed → system detects the existing review and informs the Buyer that another review cannot be submitted.
- Buyer submits a review containing prohibited content → system does not publish the review and informs the Buyer that it must be edited before resubmitting.
- Buyer submits a review without selecting a star rating → system prompts the Buyer to add a rating before the review can be submitted.
- A Buyer who has not purchased and received a given book attempts to review it → system does not display a “Write a Review” option for that Buyer on that listing.
- Buyer views a listing with existing reviews → system displays the average star rating and the individual written reviews, most recent first.

3.8.3. Functional Requirements

REQ-47: The system shall allow a Buyer to submit a star rating and a written review for a book only if that Buyer has an order containing that book which has been marked “Delivered.”

REQ-48: The system shall prevent a Buyer from submitting more than one review for the same book, and shall inform the Buyer if they attempt to do so.

REQ-49: The system shall screen every submitted review against a maintained list of prohibited words or phrases before publishing it.

REQ-50: The system shall withhold a review that fails the prohibited-content check

from publication and shall inform the Buyer that the review must be edited before it can be resubmitted.

REQ-51: The system shall require a star rating before a review can be submitted, and shall prompt the Buyer to add one if it is missing.

REQ-52: The system shall display a book's average star rating and its individual published written reviews on that book's listing detail page, ordered by submission date with the most recent first.

3.9. Wishlist

3.9.1. Description and priority

This feature allows a Buyer to save books they are interested in to a personal wishlist for later viewing, and to manage that wishlist over time, including when a saved book is no longer available. Priority: **Low**.

3.9.2. Stimulus/Response Sequences

- Buyer selects a book while searching or browsing and clicks "Add to Wishlist" → system adds the book to that Buyer's wishlist and confirms the update.
- Buyer attempts to add a book that is already saved in their wishlist → system detects the duplicate and informs the Buyer the book is already in their wishlist.
- Buyer opens their wishlist → system displays all books the Buyer has saved.
- Buyer selects a saved book and chooses "Remove from Wishlist" → system removes the book and confirms the update to the Buyer.
- A book saved in a Buyer's wishlist is no longer available (e.g., the Seller removed the listing) → system marks the saved entry as unavailable rather than deleting it outright, and the Buyer may choose to remove it.

3.9.3. Functional Requirements

REQ-53: The system shall allow a logged-in Buyer to add a book to their personal wishlist while searching or browsing.

REQ-54: The system shall detect an attempt to add a book already present in a Buyer's wishlist and shall inform the Buyer rather than creating a duplicate entry.

REQ-55: The system shall display all books saved in a Buyer's wishlist when the Buyer opens the wishlist view.

REQ-56: The system shall allow a Buyer to remove a book from their wishlist and shall confirm the update once complete.

REQ-57: The system shall detect when a book saved in a Buyer's wishlist is no longer available and shall mark it as unavailable within the wishlist rather than silently deleting it, allowing the Buyer to remove it manually if they choose.

4. Other Nonfunctional Requirements

4.1. Performance Requirements

The system shall return search and browse results within 2 seconds under normal load. The system shall complete the checkout process, from payment submission to confirmation display, within 5 seconds under normal load, excluding delays introduced by the external payment gateway itself. Cover image uploads shall complete within 5 seconds for files at or below the maximum allowed size. The admin review queue shall load within 2 seconds regardless of the number of pending listings, through the use of pagination if necessary. These targets are intended to keep the buying and selling experience responsive enough that users do not perceive the system as slow, which is important for both buyer retention and seller adoption.

4.2. Safety Requirements

BookHaven does not control physical hardware and therefore poses no direct physical safety risk. The primary safety-adjacent concern is protecting users from financial harm: the system shall never allow an order to be marked “Paid” unless the payment gateway has confirmed a successful charge, and shall never decrement a Seller’s stock or notify a Buyer of a successful purchase unless payment has actually succeeded. The system shall not permit an Administrator action (approve, reject, or refund) to be applied twice to the same listing or order as the result of a duplicate or repeated request.

4.3. Security Requirements

The system shall store all passwords using a salted cryptographic hash and shall never store or transmit passwords in plain text. All communication between the client and server, including login credentials and payment-related traffic, shall be encrypted using HTTPS/TLS. The system shall not store raw payment card numbers, in accordance with the design constraint stated in Section 2.5, relying instead on tokenized references provided by the payment gateway. The system shall enforce role-based access control so that Buyers cannot access Seller listing-management functions, Sellers cannot access another Seller’s listings, and only Administrators can access the listing review queue. The system shall log administrative actions (approvals, rejections, refunds) with a timestamp and the acting Administrator’s identity for accountability.

4.4. Software Quality Attributes

4.4.1. Availability

The system shall be available to Buyers and Sellers at least 99% of the time, excluding scheduled maintenance windows communicated in advance.

4.4.2. Dependability

The system shall behave consistently: performing the same action (e.g., adding the same listing to a cart) shall produce the same result regardless of which Buyer performs it or how many times it is repeated under identical conditions, and normal use shall not cause the system to crash or enter an unrecoverable state.

4.4.3. Usability

A first-time Buyer shall be able to search for a book and add it to their cart without external instructions, and a first-time Seller shall be able to create a listing using only the on-screen form and field labels, without consulting outside documentation.

4.4.4. Maintainability

The codebase shall be organized so that a new developer joining the team can add a new listing field (e.g., a new metadata attribute) by modifying a small, well-defined set of files, without needing to alter unrelated parts of the system.

4.4.5. Flexibility

The system shall be designed so that additional payment gateways or additional book metadata fields (e.g., language, publisher) can be added in a future release without requiring a redesign of the core account, listing, or order data models.

5. Other Requirements

Database Requirements: All persistent data (accounts, listings, orders, return requests) shall be stored in a relational database with referential integrity enforced between related tables (e.g., an order line item must reference a valid listing and a valid order).

Legal Requirements: The system shall comply with applicable consumer-protection expectations for e-commerce return windows and shall avoid storing more payment information than is necessary under PCI-DSS guidance, as noted in Section 2.5.

Internationalization: Version 1.0 of BookHaven shall support the English language and US-dollar pricing only; support for additional languages and currencies is out of scope for this release but is noted here as a likely future requirement.

Reuse Objectives: Where practical, the team shall use well-supported open-source libraries for common functionality (e.g., password hashing, ISBN validation) rather than writing custom implementations, to reduce risk and development time.

Repository and Documentation: The full project source code, commit history, and issue tracking are hosted in the team's GitHub repository, cited in reference [5]. Every functional requirement in Chapter 3 (REQ-1 through REQ-57), across all nine System Features, traces back to at least one of the nine user stories listed in Appendix A, satisfying the project requirement that all functional requirements be derived from an elicited user story.

A. User Stories

This appendix lists the user stories from which the functional requirements in Chapter 3 were derived, organized by user role and author. Each story includes preconditions, a numbered main flow, and alternate (ALT) flows describing exceptional or branching behavior.

A.1. Buyer User Stories

User Story 1 (BookHaven Team)

As a user, I want to register for an account and log in so that I can access BookHaven as a Buyer or Seller.

Precondition:

- User does not yet have a BookHaven account, or is a returning registered user

Steps:

1. User navigates to the registration page
2. User selects an account type, Buyer or Seller
3. User enters their name, a unique email address, and a password
4. User submits the registration form
5. System creates the account and logs the user in, redirecting them to their role-appropriate home page

ALT 1 — Registration is missing a required field or reuses an existing email:

- 4A System detects a missing field or that the email is already registered
- 5A User is shown an error identifying the specific problem and the account is not created

ALT 2 — Returning user logs in:

- 1B User navigates to the login page instead of the registration page
- 2B User enters their email and password
- 3B System authenticates the user and redirects them to their role-appropriate home page

ALT 3 — Login fails:

- 2C User enters an email/password combination that does not match an account on file
- 3C System displays an error message without indicating whether the email or password was incorrect, and offers a password-reset option

User Story 2 (BookHaven Team)

As a buyer, I want to search for and browse published book listings so that I can find a book I am interested in before purchasing it.

Precondition:

- At least one book listing has been published on BookHaven

Steps:

1. Buyer navigates to the BookHaven home page
2. Buyer enters a search term and selects a field to search by (title, author, genre, or ISBN), or selects a genre to browse
3. System displays all Published listings matching the search or browse criteria, sorted alphabetically by title
4. Buyer selects a listing from the results
5. System displays the full listing detail page, including description, price, condition, seller name, and stock availability

ALT 1 — Search returns no results:

- 3A System finds no Published listings matching the search term
- 4A Buyer is shown a “no results found” message and may try a different search

User Story 3 (Rakshit Jaiswal)

As a buyer, I want to search for and purchase an item so that I can complete my order.

Precondition:

- Buyer has a valid account
- Item is listed and currently in stock
- Buyer has a valid payment method on file

Steps:

1. Login
2. Buyer searches for a product using the search bar or browses by category

3. Buyer selects the desired item from the results
4. Buyer reviews item details, price, and seller information
5. Buyer clicks “Add to Cart” or “Buy Now”
6. Buyer enters a shipping address
7. Buyer selects a payment method
8. Buyer reviews the order summary
9. Buyer submits the order
10. System processes payment
11. Buyer receives an order confirmation and receipt

ALT 1 — Item goes out of stock during checkout:

- 6A System detects the item’s stock reached zero before payment is submitted
- 9A Buyer is informed the item is no longer available and prompted to remove it from the cart

ALT 2 — Payment declined:

- 9B Payment processor declines the transaction
- 10B Buyer is informed the payment failed and prompted to try another payment method

User Story 4 (Rakshit Jaiswal)

As a buyer, I want to request a return and refund for a book I purchased so that I can get my money back if it is damaged, wrong, or no longer wanted.

Precondition:

- Buyer has a completed order that includes the book
- Order is within the platform’s return eligibility window (30 days of delivery)
- Book’s condition and category are eligible for return per policy

Steps:

1. Login
2. Buyer navigates to “Order History”
3. Buyer selects the order containing the book to return
4. Buyer clicks “Request Return/Refund”

5. Buyer selects a reason for the return (damaged, wrong item, not as described, no longer wanted)
6. Buyer submits the return request with optional photos and comments
7. Seller/admin reviews the return request
8. Return request is approved
9. Buyer receives return shipping instructions and a return label
10. Buyer ships the book back to the seller
11. Seller/admin confirms receipt and condition of the returned book
12. System processes the refund to the buyer's original payment method
13. Buyer receives a refund confirmation

ALT 1 — Return window has expired:

- 3A System detects the order is outside the eligible return window
- 4A Buyer is informed the book is no longer eligible for return

ALT 2 — Return request denied:

- 8A Seller/admin reviews and denies the request (e.g., invalid reason, book not eligible)
- 9A Buyer is notified the return was denied, with a reason

ALT 3 — Returned book does not match the reported condition:

- 12A Seller/admin inspects the returned book and finds it doesn't match the reported issue (missing pages not disclosed, item swapped)
- 13A Refund is adjusted or denied per policy, and the buyer is notified with an explanation

User Story 5 (Cindy Cardona-Felix)

As a buyer, I want to write a review and rating for a book I purchased so that I can share my experience with other buyers.

Precondition:

- Buyer has a valid account
- Buyer is logged in
- Buyer has purchased the book
- The order has been delivered

- Buyer has not submitted a review for the book

Steps:

1. Buyer logs in
2. Buyer navigates to “Order History”
3. Buyer selects the completed order containing the book
4. Buyer selects “Write a Review”
5. Buyer gives the book a rating
6. Buyer writes a review
7. Buyer submits the review
8. System checks the review for prohibited content
9. System saves the review
10. Review is displayed in the reviews section for that book
11. Buyer is informed that the review was submitted successfully

ALT 1 — Buyer is attempting to review a book they already reviewed:

- 4A System detects that the buyer has already submitted a review
- 5A Buyer is informed that another review cannot be submitted

ALT 2 — Buyer submits a review with prohibited content:

- 8B System detects prohibited content in the review
- 9B System does not publish the review
- 10B Buyer is informed that the review must be edited before submitting again

ALT 3 — Buyer submits the review without a rating:

- 7C System detects that the rating is missing
- 8C Buyer is prompted to add a rating
- 9C Buyer adds the rating and submits the review again

User Story 6 (Cindy Cardona-Felix)

As a buyer, I want to manage my wishlist so that I can save books I am interested in and easily find them later.

Precondition:

- Buyer has a valid account
- Buyer is logged in
- Books are available for the buyer to view

Steps:

1. Buyer logs in
2. Buyer searches for or browses books
3. Buyer selects a book they are interested in
4. Buyer selects “Add to Wishlist”
5. System adds the book to the buyer’s wishlist
6. Buyer navigates to their wishlist
7. System displays all books saved in the wishlist
8. Buyer selects a saved book
9. Buyer chooses to remove the book or keep it saved
10. System updates the wishlist
11. Buyer is informed that the wishlist was updated successfully

ALT 1 — Buyer attempts to add a book that is already in the wishlist:

- 4A System detects that the book is already saved
- 5A Buyer is informed that the book is already in their wishlist

ALT 2 — Buyer removes a book from the wishlist:

- 9B Buyer selects “Remove from Wishlist”
- 10B System removes the book from the wishlist
- 11B Buyer is informed that the book was removed successfully

ALT 3 — A saved book is no longer available:

- 7C System detects that a saved book is no longer available
- 8C System marks the book as unavailable
- 9C Buyer can choose to remove the unavailable book from the wishlist

User Story 7 (Jalil Jimenez)

As a buyer, I want to edit my account information so that my profile stays accurate over time.

Precondition:

- Buyer has a valid account and is logged in

Steps:

1. Buyer navigates to “Account Settings”
2. Buyer selects the field(s) to update (name, email, or password)
3. Buyer enters the new information
4. Buyer submits the changes
5. System validates the new information
6. System saves the updated account information
7. Buyer is shown a confirmation that their account was updated

ALT 1 — Buyer enters an email already used by another account:

- 5A System detects the email is already associated with an existing account
- 6A Buyer is shown an error and the change is not saved

ALT 2 — Buyer updates their password:

- 3B Buyer enters their current password, followed by a new password and confirmation
- 5B System verifies the current password is correct and the new password meets complexity requirements
- 6B System updates the password and ends all other active sessions for the account

ALT 3 — Buyer attempts to change their account type:

- 2C Buyer attempts to switch from Buyer to Seller in the account settings form
- 3C System does not display account type as an editable field, since account type cannot be changed after registration

User Story 8 (Jalil Jimenez)

As a buyer, I want to remove items from my shopping cart and see the subtotal update so that I can adjust my order before checking out.

Precondition:

- Buyer is logged in

- Buyer has at least one item currently in their cart

Steps:

1. Buyer navigates to “My Cart”
2. System displays each item in the cart along with its price and the current subtotal
3. Buyer selects “Remove” on an item they no longer want
4. System removes the item from the cart
5. System recalculates and displays the updated subtotal
6. Buyer either continues shopping or proceeds to checkout with the remaining items

ALT 1 — Buyer removes the last item in their cart:

- 4A System detects the cart is now empty
- 5A System displays an empty-cart message and hides the “Checkout” option until an item is added

ALT 2 — Item in the cart goes out of stock before removal:

- 2B System detects that an item in the cart is now out of stock
- 3B System labels the item as unavailable within the cart and prompts the Buyer to remove it before checking out

A.2. Seller User Stories

User Story 9 (Everett Wappler)

As a seller, I want to list a physical/digital item detailing product name, photo, description, a price, inventory, and contact information.

Precondition:

- Seller has a valid account
- Seller has items for sale
- Seller has all information ready

Steps:

1. Login
2. Seller navigates to My Listings
3. Seller clicks Add New Listing

4. Seller selects option Physical
5. Seller enters valid information into field for Product Name
6. Seller enters valid ISBN
7. Seller selects an option for Genre
8. Seller selects an option for Item Condition
9. Seller enters valid information into field for Description
10. Seller enters valid information into field for Contact Information
11. Seller enters valid information into field for Inventory
12. Seller enters valid information into field for Price
13. Seller uploads valid Picture of allowed size and file type
14. Seller selects a shipping option and specifies handling time
15. Seller submits Review Order
16. Seller submits the product creation form
17. Seller is informed and Admin is pending approval

ALT 1 — Seller lists a digital item for sale:

- 4A Seller selects option Digital
- 8A Item Condition is automatically set to N/A
- 11A Inventory is automatically set to N/A
- 14A Seller uploads a valid source manuscript file
- 15A Seller selects a licensing/DRM option
- 16A Seller sets any usage limits such as device count, download limit, or expiration
- 17A Seller is informed and Admin is pending approval

ALT 2 — Seller has missing required fields while listing a product:

- 15B System detects a required field such as price, file, or inventory is missing at review
- 16B Seller is prompted to complete the missing fields before the listing can be submitted for review

User Story 10 (Everett Wappler)

As a seller, I want to fulfill a buyer's order so that the buyer receives their book and the sale is completed.

Precondition:

- Seller has a valid account
- Seller has at least one order in "Pending Orders" status
- Seller has a valid shipping/carrier option
- Buyer's payment has been successfully processed

Steps:

1. Login
2. Seller navigates to Pending Orders
3. Seller selects Process Order from list of pending orders
4. Seller reviews order details from the buyer
5. Seller marks the order as "Processing"
6. Seller selects a shipping carrier
7. Seller enters the valid tracking information
8. Seller submits Review Order
9. Seller submits the pending order submission form
10. Buyer is notified that their order has been shipped with the tracking information

ALT 1 — Seller cannot fulfill the order:

- 3A Seller selects Cancel Order from list of pending orders
- 4A Seller enters required information for reason of cancellation
- 5A Buyer is notified and refunded

ALT 2 — System fulfills an order for a digital book:

- 2A System detects payment confirmation for a digital order
- 3B System automatically generates a unique access token or license tied to the buyer's account
- 4B System delivers the file or link to the buyer

5B Seller receives a notification that their digital order was fulfilled automatically

User Story 11 (Jalil Jimenez)

As a seller, I want to edit or remove an existing book listing so that my inventory stays accurate and up to date.

Precondition:

- Seller has a valid account and is logged in
- Seller has at least one existing listing, published or pending

Steps:

1. Seller navigates to “My Listings”
2. Seller selects an existing listing
3. Seller chooses to either “Edit” or “Remove” the listing
4. If editing, Seller updates one or more fields (price, description, stock, cover image, etc.)
5. Seller submits the changes
6. System saves the updated listing information
7. Seller is shown a confirmation that the listing was updated

ALT 1 — Seller edits a field relevant to moderation:

- 4A Seller changes the title, description, or ISBN of an already-published listing
- 6A System returns the listing to “Pending Review” and temporarily hides it from Buyers until an Administrator re-approves it

ALT 2 — Seller removes a listing instead of editing it:

- 3B Seller selects “Remove” instead of “Edit”
- 4B System asks the Seller to confirm the removal
- 5B Seller confirms
- 6B System deletes the listing from the store, regardless of its prior review status

ALT 3 — Seller attempts to edit another seller’s listing:

- 2C Seller attempts to access a listing that does not belong to them (e.g., via a direct link)
- 3C System denies access and displays an error indicating the Seller does not have permission to modify that listing

A.3. Administrator User Stories

User Story 12 (BookHaven Team)

As an admin, I want to review and approve or reject pending book listings so that only appropriate, accurate listings are published on the site.

Precondition:

- Admin has a valid administrative account
- At least one book listing is pending approval
- Admin has reviewed the platform's content policies

Steps:

1. Login (see User Story 1)
2. Click side tab, select "Pending Listings"
3. Admin selects a pending book listing to review
4. Admin reviews the title, author, ISBN, genre, condition, description, cover image, price, and stock for policy compliance
5. Admin verifies the ISBN is valid and checks for blacklisted words/phrases or prohibited images
6. Admin approves the listing, or flags/rejects it
7. If approved, the book becomes publicly visible in the store
8. If rejected, the seller is notified with a reason

ALT 1 — Listing contains prohibited content or an invalid ISBN:

- 5A Admin identifies a blacklisted word/phrase, inappropriate image, or invalid ISBN
- 6A Admin rejects the listing and selects a reason code
- 8A Seller is notified their book listing was rejected, with the reason provided

ALT 2 — Listing needs minor seller correction:

- 5B Admin identifies a minor issue that does not require outright rejection
- 6B Admin returns the listing to the seller with comments requesting changes
- 8B Seller is informed to update and resubmit the listing

User Story 13 (Jalil Jimenez)

As an admin, I want to maintain the list of prohibited words and phrases so that book listings and reviews are automatically screened for inappropriate content.

Precondition:

- Admin has a valid administrative account and is logged in
- Admin has permission to manage content-moderation settings

Steps:

1. Admin navigates to “Content Moderation Settings”
2. System displays the current list of prohibited words and phrases
3. Admin adds a new word or phrase to the list
4. Admin submits the change
5. System validates that the entry is not already present
6. System saves the updated list
7. System immediately applies the updated list to future listing and review submissions

ALT 1 — Admin attempts to add a duplicate entry:

- 5A System detects the word or phrase is already on the list
- 6A Admin is informed the entry already exists and no duplicate is added

ALT 2 — Admin removes an entry from the list:

- 3B Admin selects “Remove” next to an existing entry instead of adding a new one
- 4B System asks the Admin to confirm the removal
- 5B Admin confirms
- 6B System removes the entry and applies the change to future submissions

ALT 3 — Admin needs to reprocess already-pending content against the updated list:

- 7C Admin selects “Re-scan Pending Listings” after updating the list
- 8C System re-checks all currently pending listings against the updated prohibited-content list and flags any new matches for Admin attention

Point of Contact

For further information regarding this document or the BookHaven project, please contact any member of the development team: Cindy Cardona-Felix, Rakshit Jaiswal, Jalil Jimenez, or Everett Wappler. The team's source code, issue tracker, and supporting project documentation are hosted at the GitHub repository cited in reference [5]. All content in this document reflects the team's own design decisions for the BookHaven project and is provided for course evaluation purposes.